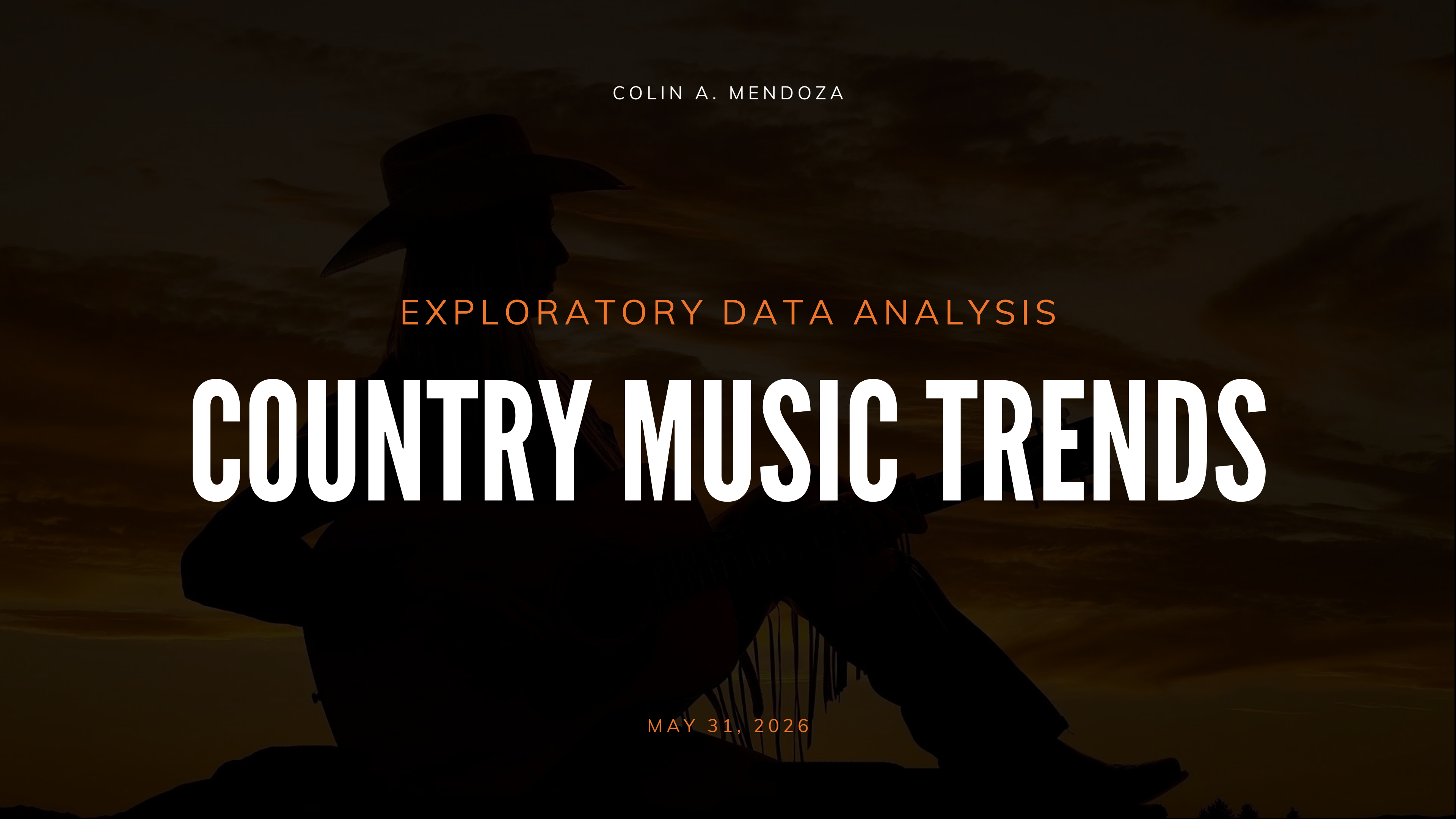




COLIN A. MENDOZA

EXPLORATORY DATA ANALYSIS

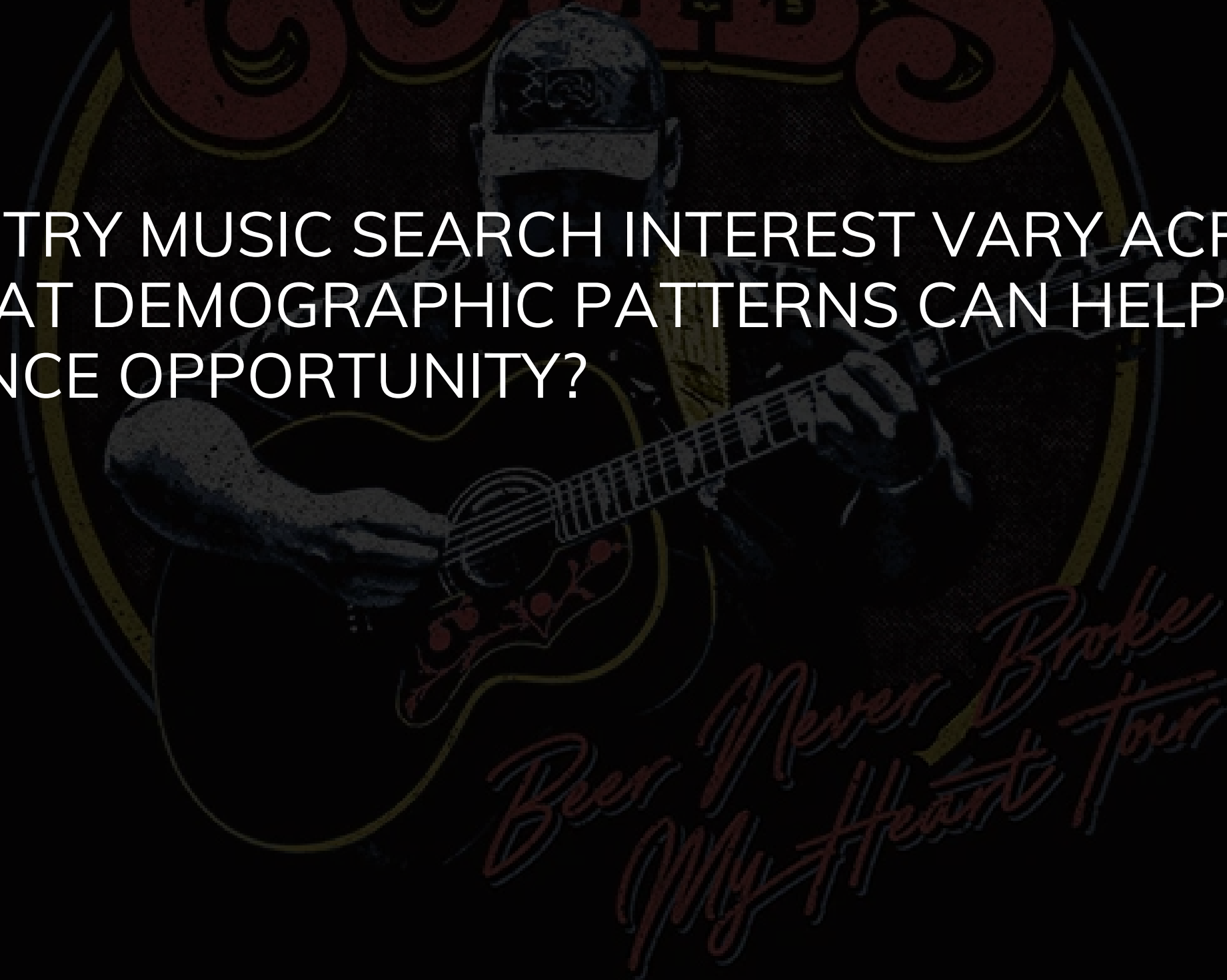
COUNTRY MUSIC TRENDS

MAY 31, 2026

PROJECT/BUSINESS QUESTION

EXPLORATORY DATA ANALYSIS

HOW DOES COUNTRY MUSIC SEARCH INTEREST VARY ACROSS U.S. STATES, AND WHAT DEMOGRAPHIC PATTERNS CAN HELP EXPLAIN REGIONAL AUDIENCE OPPORTUNITY?



DATASETS

EXPLORATORY DATA ANALYSIS

TABLE 1



Google Trends / pytrends
Country Music Interest
Scale: 1 - 100

TABLE 2



U.S. Census Bureau
State Demographics
Education, Race, Income

DATA CLEANING & ORGANIZING

EXPLORATORY DATA ANALYSIS



JOINS

Used an INNER JOIN to put together search trends table and demographic table, and joining ON state.



STANDARDIZED STATES TO ABBREV.

Updated table by setting a CASE statement WHEN '(State Name)' THEN '(State Abbrev.)' for each state



ADDED REGION COLUMN FOR GROUPING

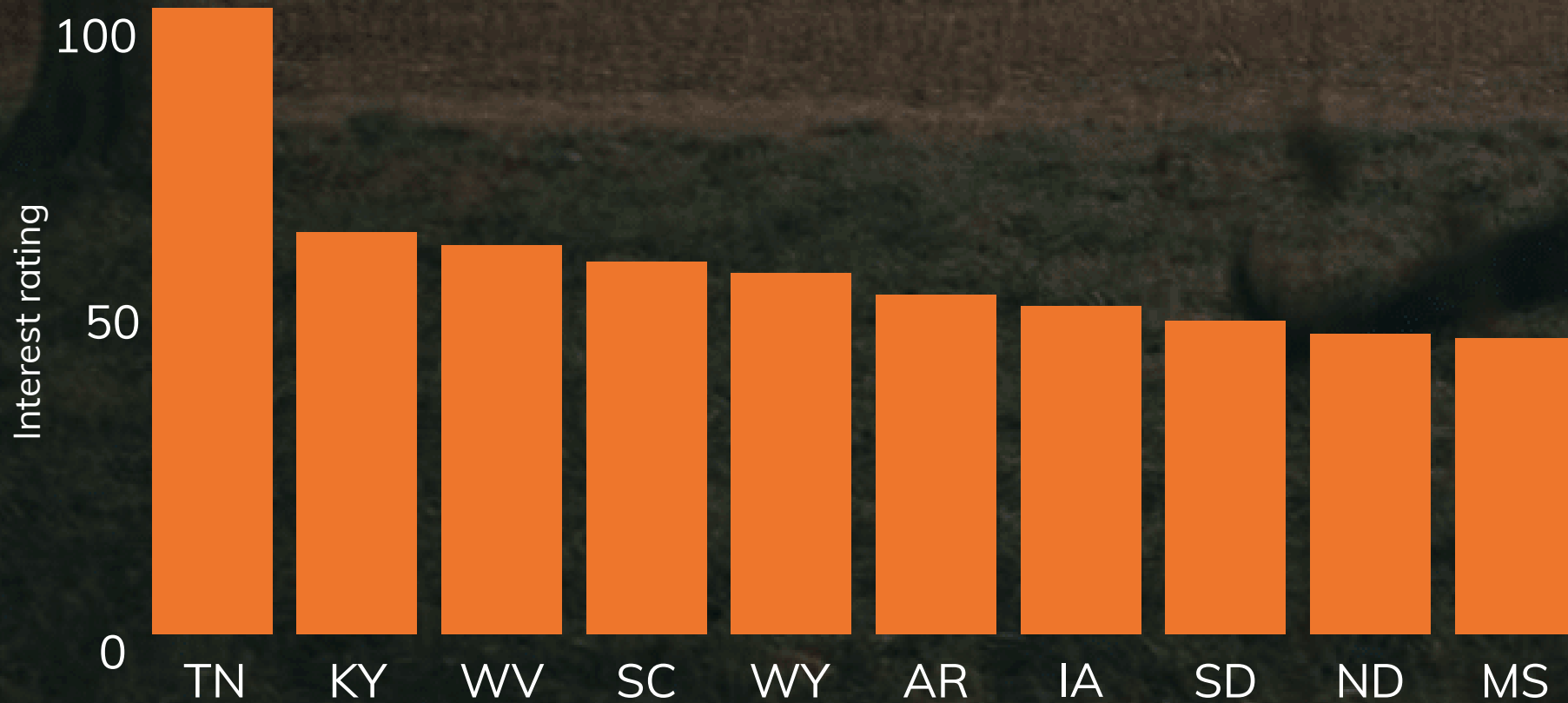
Altered table through adding a region column by setting a CASE statement WHEN '(State Name)' THEN '(State Region)' for each state

TOP AND BOTTOM 10 STATES

EXPLORATORY DATA ANALYSIS

TOP 10 STATES

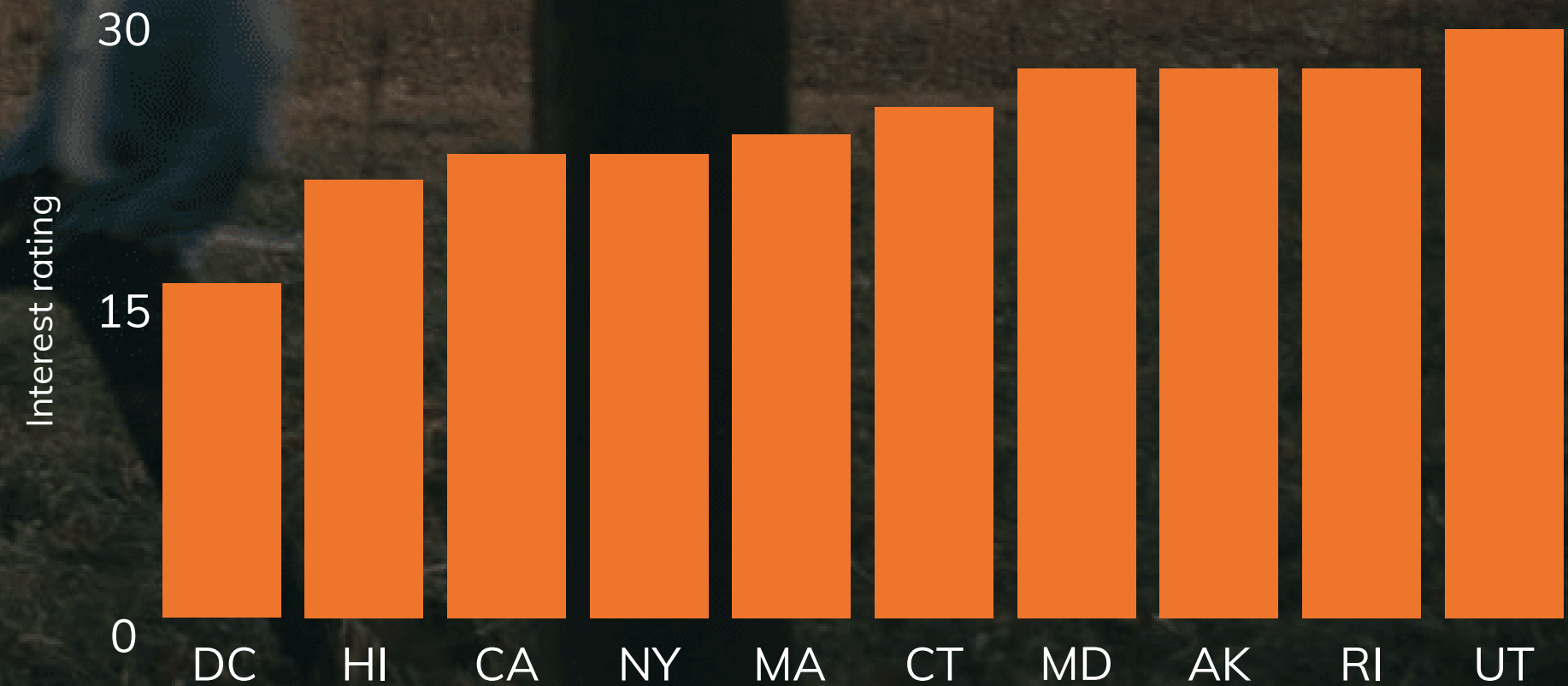
HIGH INTEREST



NORTH: 0
WEST: 1
MIDWEST: 3
SOUTH: 6

BOTTOM 10 STATES

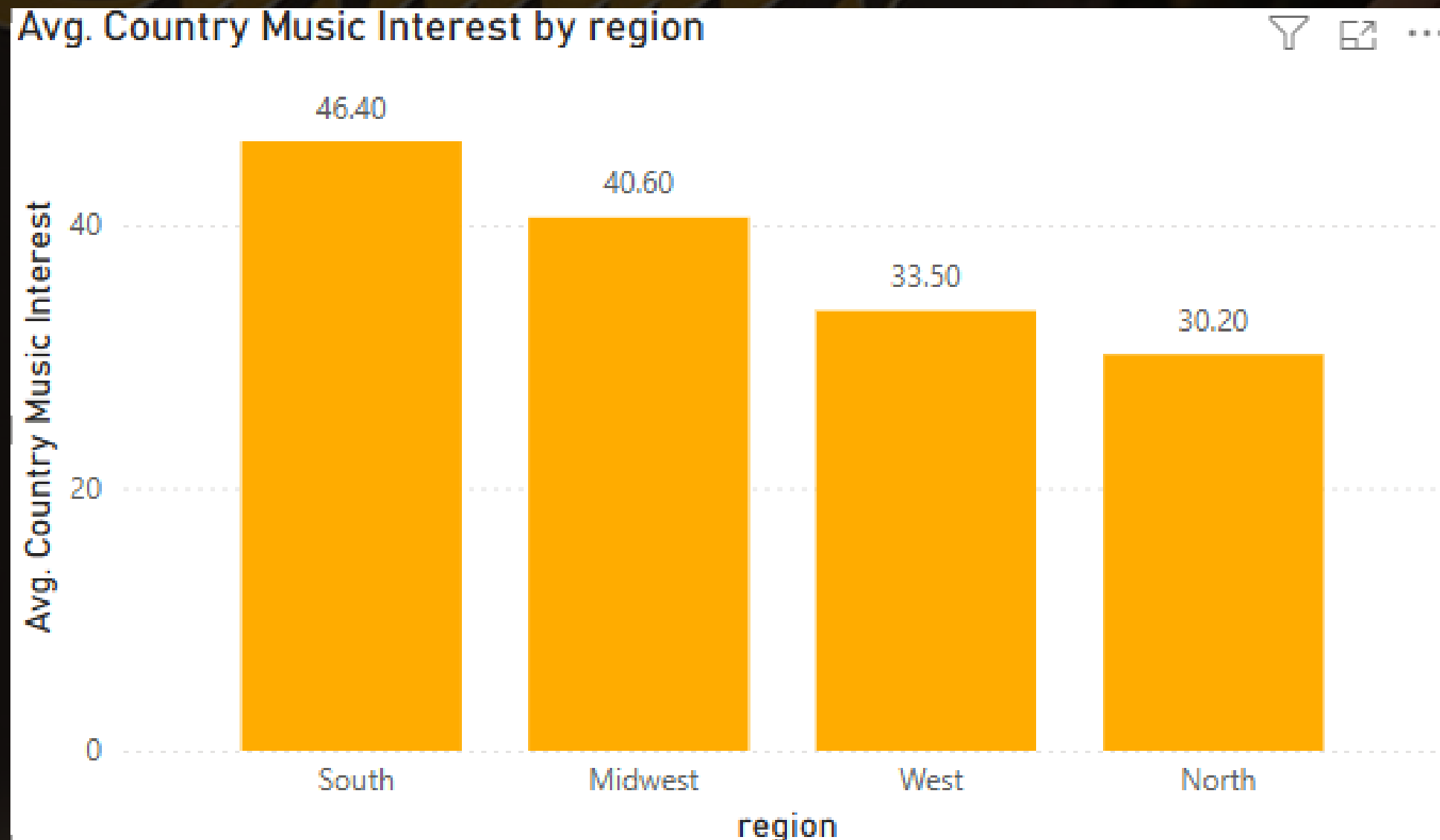
LOW INTEREST



NORTH: 5
WEST: 5
MIDWEST: 0
SOUTH: 0

AVERAGE COUNTRY INTEREST BY REGION

EXPLORATORY DATA ANALYSIS



GRAPH DESCRIPTION

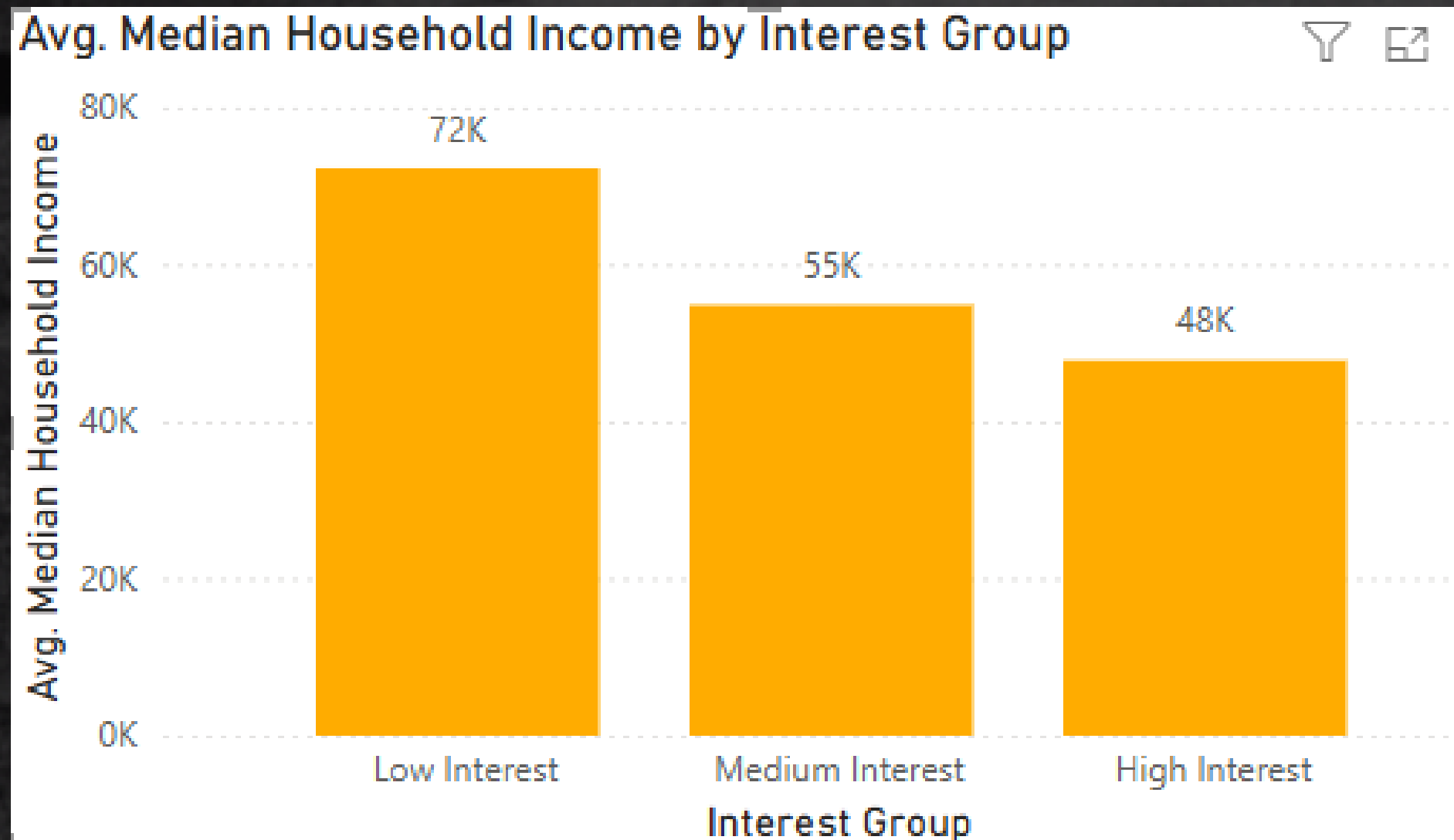
This chart compares the average country music search interest across the South, North, Midwest, and West.

INSIGHTS

Although the South leads, the Midwest/North shows meaningful interest, suggesting that country music's audience reach extends beyond the South.

DEMOGRAPHICS

EXPLORATORY DATA ANALYSIS



GRAPH DESCRIPTION

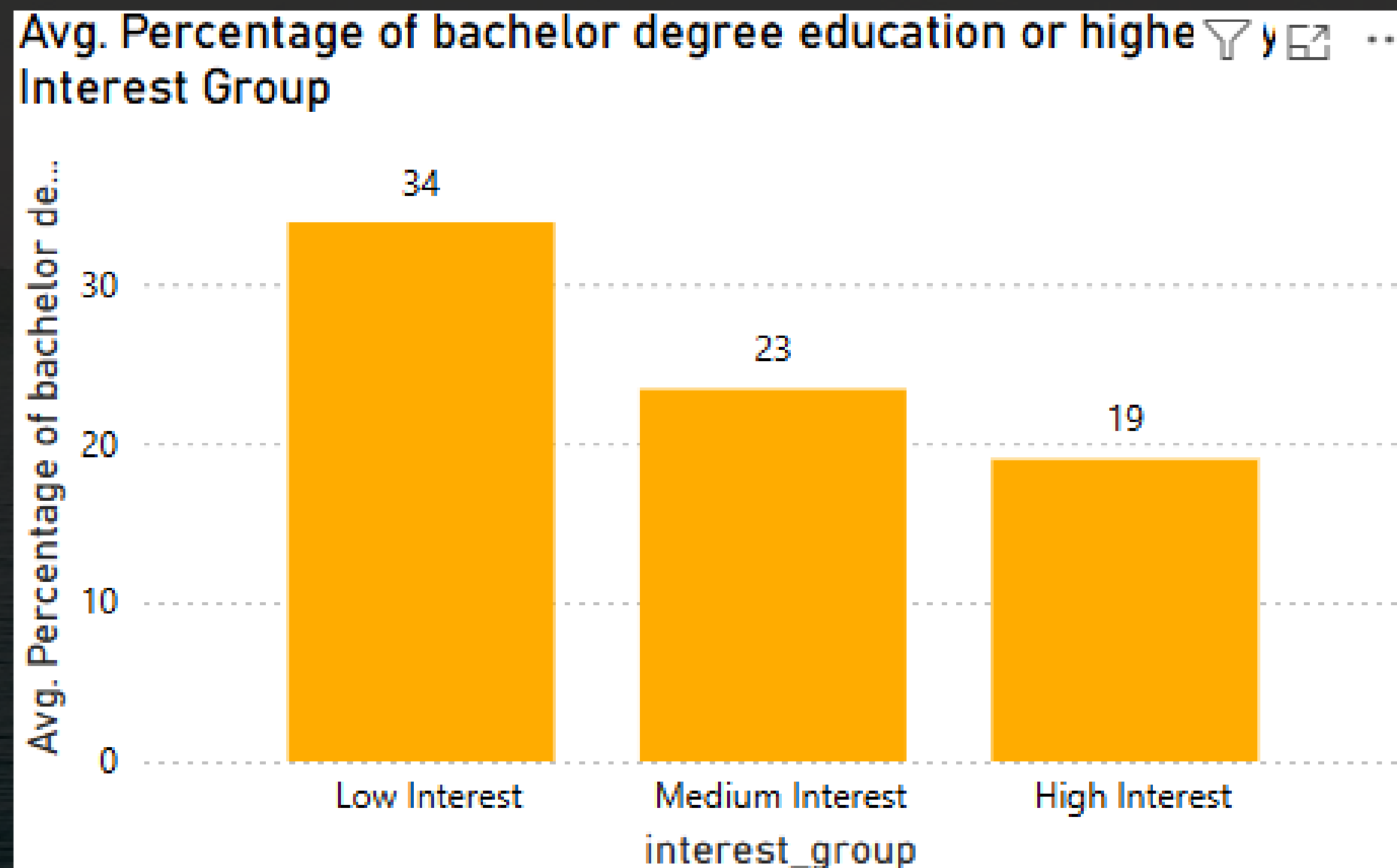
This chart compares average median household income across high, medium, and low country music interest groups.

INSIGHTS

- The low interest group has the highest average income.
- The high interest group has the lowest average income.
- This suggests country music interest may be associated with differences in state-level income

DEMOGRAPHICS

EXPLORATORY DATA ANALYSIS



GRAPH DESCRIPTION

This chart compares the average percentage of residents with a bachelor's degree or higher across country music interest groups.

INSIGHTS

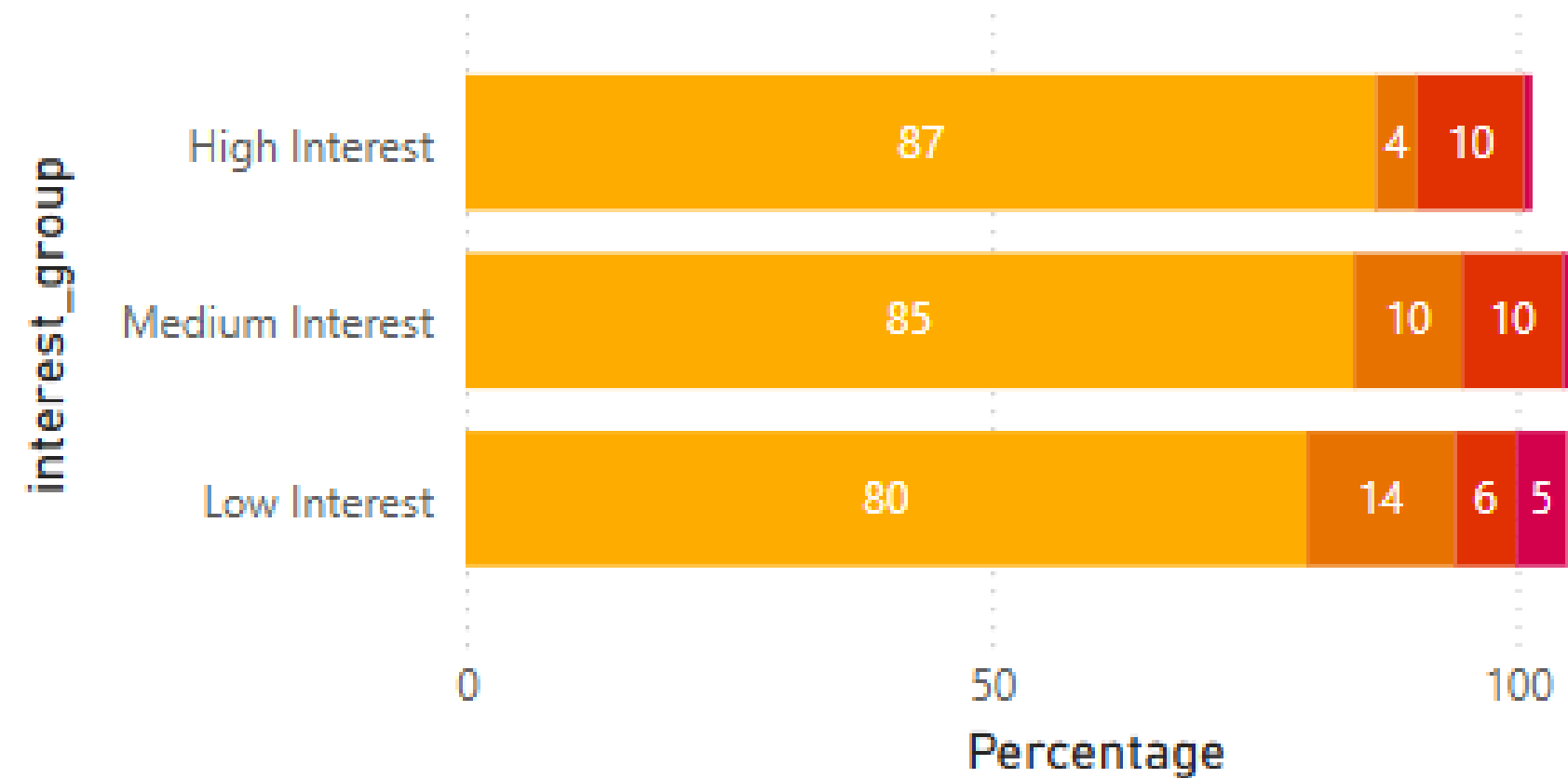
- The low interest group has the highest bachelor's degree percentage.
- The high] interest group has the lowest bachelor's degree percentage.
- This helps show that states with lower level of interest in country music have populations with more people with bachelor degrees or higher

DEMOGRAPHICS

EXPLORATORY DATA ANALYSIS

Ethnicity Breakdown by Interest Group

● avg_white_p ● avg_hispanic_or_latino_p ● avg_black_p ● avg_asian_p



GRAPH DESCRIPTION

This chart takes every state that falls within its interest group, and averages the states ethnic percentages.

INSIGHTS

- High-interest states appear to have higher average White and Black percentages.
- Lower interest states appear to have higher average Hispanic and Asian percentages.
- Important: this describes state demographics, not the exact demographics of country music listeners.

KEY FINDINGS

EXPLORATORY DATA ANALYSIS

- Country music search interest is strongest in the South, with an average interest score of 46.4, making it the clearest regional market for country music demand.
- The Midwest has the second-highest average interest score at 40.6, showing that country music audience opportunity is not limited to the South.
- The top 10 highest-interest states are heavily concentrated in the South and Midwest, with 6 Southern states and 3 Midwestern states appearing in the top group.
- The lowest-interest states are concentrated in the North and West, with 5 Northern states and 5 Western states appearing in the bottom 10.
- Demographic patterns show that high-interest states have lower average median household income at about \$48K, while low-interest states average about \$72K.
- Education follows a similar pattern: high-interest states average 19% bachelor's degree or higher, compared with 34% in low-interest states.
- Ethnicity data shows high-interest states have higher average White and Black population percentages, while low-interest states have higher average Hispanic/Latino and Asian percentages.

BUSINESS RECOMMENDATIONS

EXPLORATORY DATA ANALYSIS

- Prioritize Southern states for core country music marketing because they show the highest average search interest and contain most of the top-performing states.
- Treat the Midwest as a strong secondary market because its average interest score is close to the South, suggesting meaningful country music demand outside the genre's traditional base.
- Use Northern and Western states as selective growth markets rather than primary targets, since they show lower average interest but still contain states with measurable demand.
- In high-interest states, marketing campaigns should focus on fan retention, tour promotion, radio/playlist support, and artist engagement, because existing demand is already strong.
- In medium-interest states, campaigns should focus on audience growth, especially through crossover artists, social media, streaming playlists, and regional touring.
- Since high-interest states have lower average income, pricing strategy matters: music companies should consider affordable ticket tiers, regional merch pricing, and accessible fan experiences in stronger country markets.
- Because lower-interest states have higher income and education levels, music companies could test more targeted campaigns around premium live experiences, festival packages, or crossover country artists in those markets.

LIMITATIONS

EXPLORATORY DATA ANALYSIS

- Google Trends measures relative search interest, not exact country music listeners, streams, ticket sales, or revenue.
- The demographic data is measured at the state level, so the analysis cannot prove the exact demographics of individual country music listeners.
- The project shows patterns between search interest and demographics, but it does not prove causation.
- The income, education, and ethnicity patterns describe the states where interest is higher or lower, not the personal background of every listener in those states.

JASON ALDEAN

Backroads Of My Memory

SPOTIFY RECOMMENDATIONS

EXPLORATORY DATA ANALYSIS

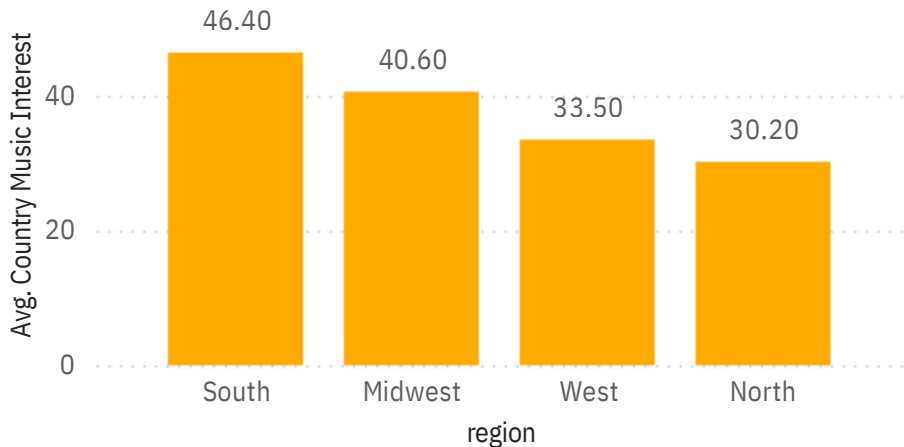
- **Audience Segmentation:**
- Spotify has tools like Spotify for Artists, where artists can understand their audience segments and listener behavior. Spotify's own guide says audience segments help artists "understand your listeners better" and make informed decisions to reach their goals.
- **Regional Demand:**
- Spotify needs to understand where genres, artists, and campaigns are gaining traction. The state map and regional analysis identify where country music interest is strongest and where potential growth markets may exist.
- **Recommendations & Discovery:**
- Spotify uses signals like streams, saves, skips, playlist adds, follows, and location to recommend content. This project uses search interest as a public-data signal to estimate where country music demand may be strongest.
- **Artist & Advertising Strategy:**
- The analysis can help identify core markets, growth markets, and lower-interest markets for country music promotion. With Spotify's first-party data, this same approach could support playlist strategy, artist marketing, regional ad targeting, and fan-growth campaigns.

JASON ALDEAN Backroads Of My Memory

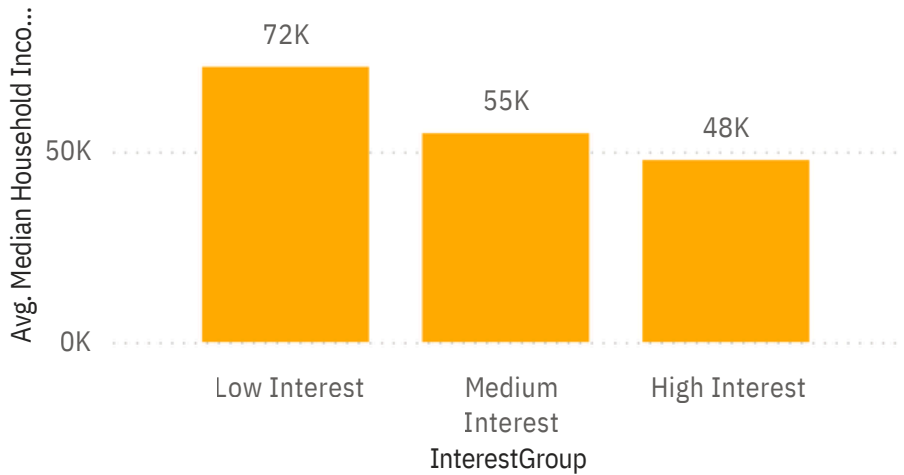
FULL DASHBOARD

EXPLORATORY DATA ANALYSIS

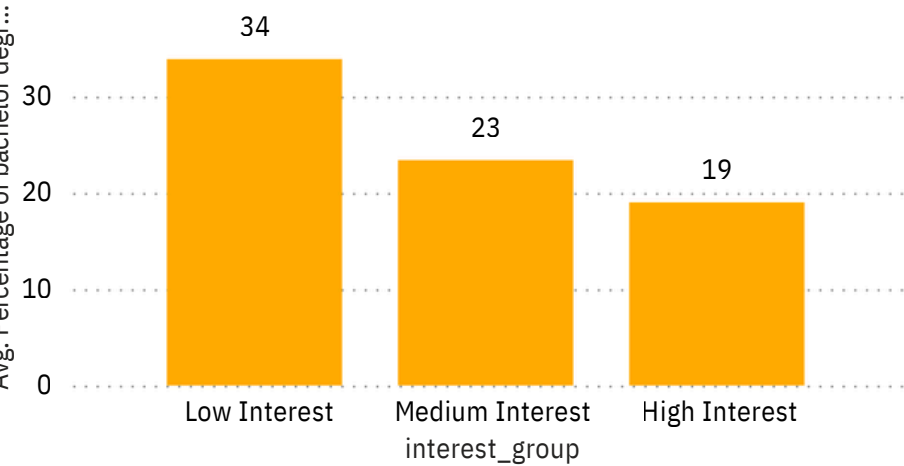
Avg. Country Music Interest by region



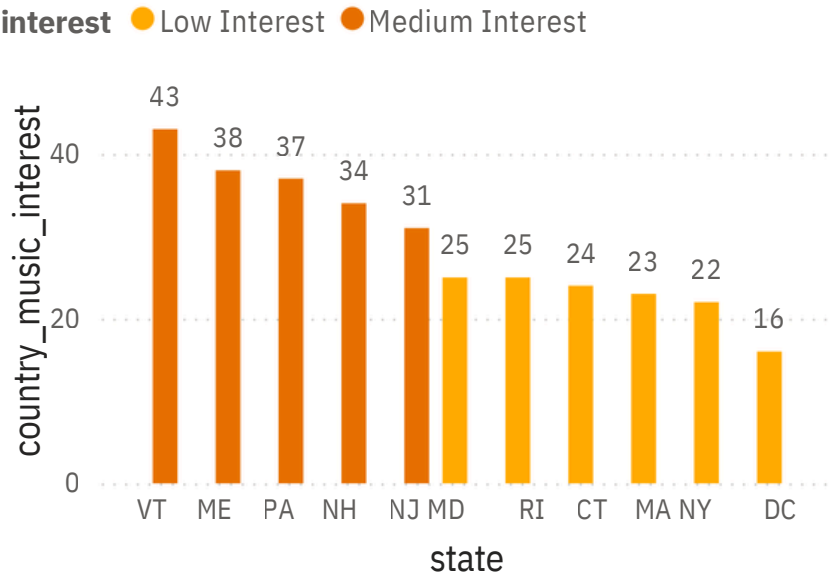
Avg. Median Household Income by Interest Group



Avg. Percentage of bachelor degree education or higher by Interest Group

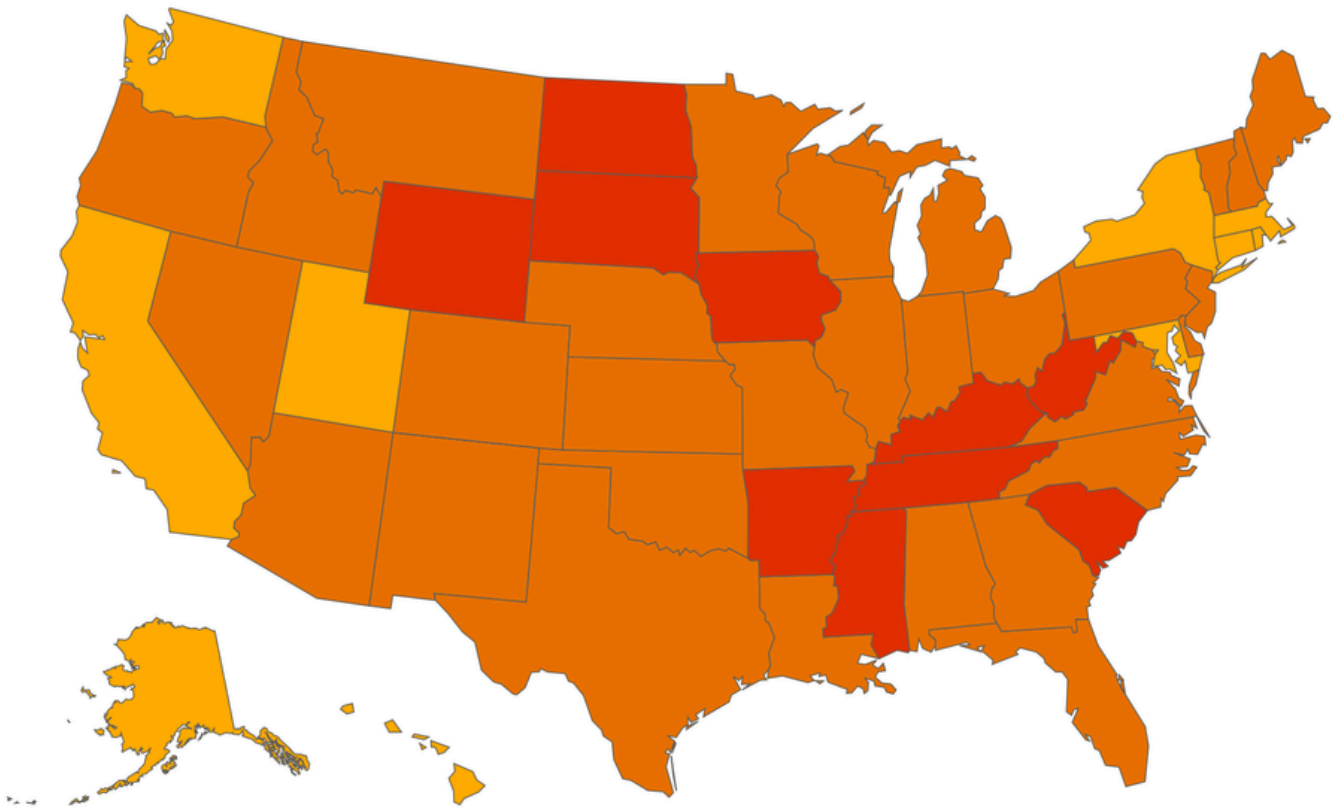


Northern States Focus



state and interest_group

Low Interest Medium Interest High Interest



Ethnicity Breakdown by Interest Group

